Answers

The sheet with its answers, **in blue**.

Part 1 — Ringville

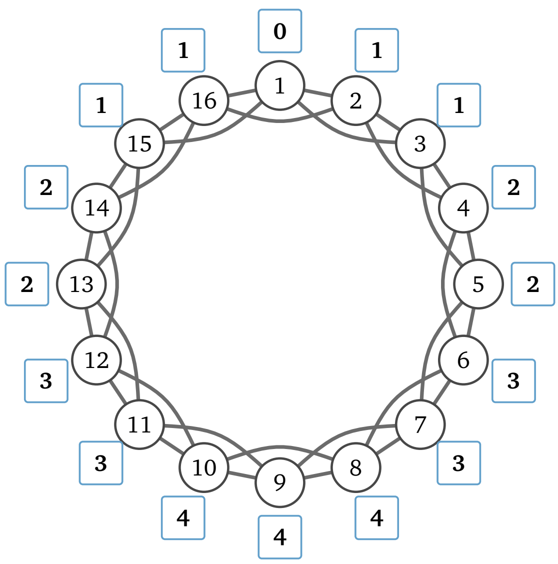
Sixteen people live in Ringville and they sit in one circle. Everybody is friends with the two on their left and the two on their right: four friends each, thirty-two friendships in the town.

Question 1(a): Person 1 has a letter for person 9. Trace the chain on the drawing. How many handshakes does it take?

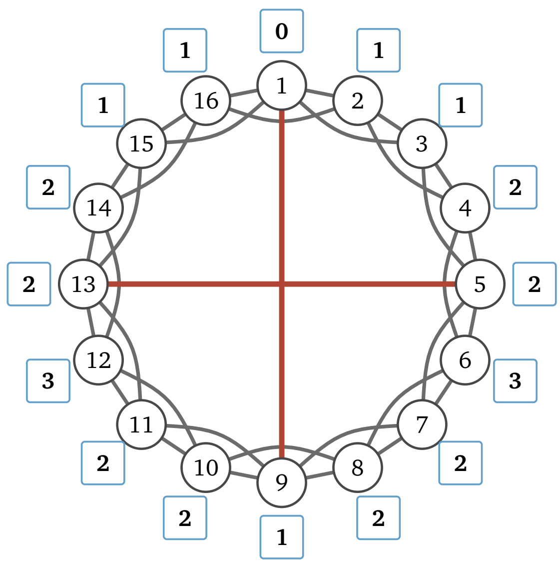
⇒ **Four**, and there are exactly two chains that do it: **1–3–5–7–9** and **1–15–13–11–9**. Person 9 is eight chairs away and no single handshake covers more than two chairs, so nothing beats four.

Question 1(b): Now everybody. In each person's box on the drawing, write how many handshakes it takes to reach them from person 1. Two are written in already, and you may find you only have to work out half of the rest.

⇒ Below left. Going round one way gives the same numbers as going round the other, so people 10 to 16 are people 8 to 2 read backwards — half the work, and the reason the drawing is symmetric is that person 1 has no idea which way is which.



1(b): Ringville



3: with the two shortcuts

Question 1(c): Add up every number in the boxes. What is the total? Share that total out among the fifteen other people — the town's average. And what is the biggest number in any box, the town's worst case?

⇒ Total **36**, average $36/15 = $ **2.4**, worst case **4**, which three people share: 8, 9 and 10.

Question 1(d): A newspaper wants one number for a headline; the post office wants one number for a promise printed on the postbox, "every letter arrives in X handshakes or fewer." Which of the two numbers goes where, and what goes wrong if you swap them?

⇒ **2.4 is the headline** and **4 goes on the postbox**. An average describes the letter you would typically send and promises nothing about any one letter: print 2.4 on the postbox and every letter that takes three or four handshakes breaks the promise. The worst case is the only number a promise can stand on, because no letter in this town ever takes longer — and it is a poor headline, because it describes three people out of sixteen. Both numbers get names in the lecture: **average path length** and **diameter**.

Question 2: Ringville grows. A thousand people, still one circle, still four friends each. How many handshakes does it take to reach the person sitting directly opposite you? And if the circle held all eight billion people on Earth? Could a world wired like Ringville pass a letter from Omaha to Boston in six handshakes?

⇒ With a thousand people the person opposite is 500 chairs away, so **250** handshakes. With eight billion they are four billion chairs away, so **two billion**. **No** — in a ring the distance grows in step with the town, so a ring-wired world would need billions of handshakes, not six. Whatever explains Milgram, it is not this shape.

Part 2 — Two shortcuts

Two of the sixteen went to college and came home with a friend from the far side of town: person 1 is now also friends with person 9, and person 5 with person 13 — thirty-four friendships, not thirty-two.

Question 3: Write the handshake counts into the boxes again, this time for the town with the two new friendships. Then the same three numbers as in 1(c): the new total, the new average, and the new worst case.

⇒ Above right. Total **27**, average $27/15 =$ **1.8**, worst case **3**, now reached only by persons 6 and 12.

Question 4: Thirty-two friendships became thirty-four. Work out both changes as percentages: by how much did the number of friendships change, and by how much did the average change? Say what those two new friendships did that thirty-two ordinary ones could not.

⇒ Friendships up by $2/32 =$ **6.25%**; the average down from 2.4 to 1.8, which is **25%**. Four times the effect, for one sixteenth of the wiring.

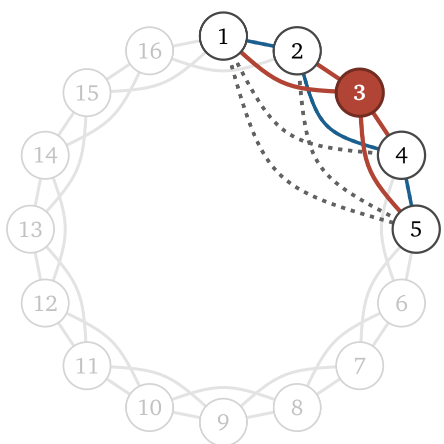
An ordinary Ringville friendship joins two people who were already one or two handshakes apart, so it shortens nothing for anybody. These two joined people *four* handshakes apart — and the saving is not theirs alone. Once person 1 can reach person 9 in a single step, so can everybody sitting near person 1, for everybody sitting near person 9. One line rewrote the distance between two whole neighbourhoods.

Part 3 — Do your friends know each other?

Ringville again, as it was in Part 1, before the two shortcuts. Person 3 is in red and their four friends are persons 1, 2, 4 and 5. Those four can be paired up six ways, and the six pairs are the dashed lines.

Darken the dashed lines that are real friendships.

⇒ **Three of the six**, drawn below in blue: **1 & 2**, **2 & 4** and **4 & 5**. The other three — 1 & 4, 1 & 5, 2 & 5 — sit three or four chairs apart, which is further than a Ringville friendship reaches.



Question 5(a): Ringville grows to ten thousand people, still four friends each. How many of person 3's six pairs are friends now?

⇒ **Still three of six.** Person 3's friends are still the people in chairs 1, 2, 4 and 5, and whether two of them are friends depends only on how many chairs apart they sit. The size of the town never enters the question.

Question 5(b): A different town: the same sixteen people and the same thirty-two friendships, but this time each friendship is drawn between two people uniformly at random. Pick two people. What is the chance that they are friends? So how many of person 3's six pairs would you expect to be friends? And in a random town of ten thousand, still four friends each?

$\Rightarrow$ Each person has 4 friends among the 15 others, so the chance is $4/15 \approx$ **0.27**, and $6 \times 4/15 =$ **1.6** of the six pairs. In a random town of ten thousand it is $4/9999$, about one in 2,500, so $6 \times 4/9999 =$ **0.0024 — effectively none**.

Question 5(c): The number of person 3's six pairs that really are friendships, divided by six, is person 3's **local clustering**. Compare the four numbers. What is it about the way friends are made in each of the two towns that decides whether local clustering survives when the town grows?

$\Rightarrow$ In Ringville your friends are **the people sitting next to you**, so they sit next to each other too, and that stays true however many chairs the circle has: **$1/2$ at any size**. In the random town your friends are **drawn from everybody**, so the chance that two of them happened to draw each other **falls away as $1/n$**. Clustering survives growth when friendships are made locally and dies when they are made globally. That is the whole reason the two halves of this module are a puzzle at all: **short paths come free from randomness, and high clustering does not**.

Three people who are all friends with each other make a **triangle**. Ringville, before the two shortcuts, has sixteen.

Question 6(a): Now go back to the drawing in Part 2, where the two shortcuts have gone in. Do persons 1 and 9 have a friend in common? Do persons 5 and 13? So how many triangles does the town have now?

$\Rightarrow$ Person 1's friends are 15, 16, 2, 3; person 9's are 7, 8, 10, 11 — **no overlap**, and **none** for 5 and 13 either, so neither new line closes a triangle. And a line that is *added* can never destroy one. So **sixteen, unchanged**. The average fell by a quarter and the town lost nothing.

Question 6(b): In Part 2 the two shortcuts were new friendships. The real model never adds one: it takes a friendship away and puts it back somewhere else. Do that once — person 1 stops being friends with person 2, and becomes friends with person 9 instead. Which friends did persons 1 and 2 have in common? So how many triangles does the town lose?

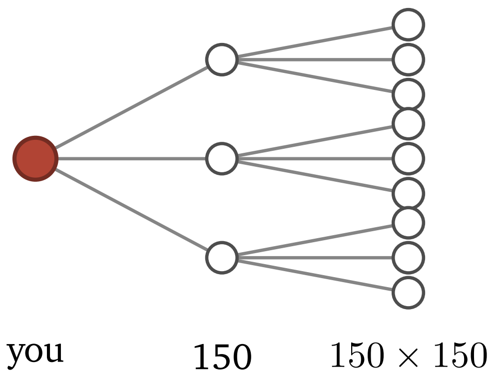
$\Rightarrow$ Persons 1 and 2 both know **person 16 and person 3**, so cutting 1–2 kills the triangles **1–2–16** and **1–2–3**: **two** go, and the town is left with **fourteen**. This is what the Watts–Strogatz model actually does at each step, and it is the honest version of 6(a).

Question 6(c): What does a shortcut buy the town, and what does it cost?

$\Rightarrow$ It buys **a quarter off the average distance** for two lines out of thirty-four. It costs **nothing when the line is added, and two triangles out of sixteen when it is moved** — and the triangles it breaks are all local to the one person it was moved from. **Distance is cheap to buy and clustering is expensive to lose**, so a handful of shortcuts collapses the distances and leaves almost all of the local structure standing. That is the mechanism the whole model rests on.

Part 4 — Six handshakes?

Question 7(a): Everybody knows 150 people. Pretend for a moment that no two of your friends know each other, so every handshake reaches 150 people nobody in the chain has met yet. Carry the table two rows further. At how many handshakes do you pass eight billion?



the branching, drawn with 3 friends each instead of 150

handshakes	people reached
1	150
2	22,500
3	about 3.4 million
4	**about 506 million**
5	**about 76 billion**

⇒ **Five.** Four handshakes reach half a billion, which is not enough; five overshoot the planet by a factor of nine. Six degrees was never a measurement, it was this multiplication.

Question 7(b): Every branch of the tree lands on somebody new, so no person appears in it twice. But in Part 3 you found that your friends are often friends with each other, so some branches must land on a person who is already in the tree. In a real town, is the true number of handshakes bigger or smaller than the one you counted in 7(a)? Why?

⇒ **Bigger — more handshakes than five.** Half of person 3's pairs of friends are friends with each other, so most of the 150 people your friend reaches are people you were already reaching yourself. Those branches land back inside the tree instead of on somebody new, so every step adds far fewer than 150 times as many people. The table is therefore an **overestimate of the reach** and an **underestimate of the number of handshakes**. Clustering is exactly what makes six degrees hard.

Two functions and one open-ended cell. Everything below except the lines marked <-- is already on their screen.

```
def ring_edges(n, half):
    edges = []
    for i in range(n):
        for d in range(1, half + 1):
            j = (i + d) % n                 <--
            edges.append((i, j))
    return edges

def local_clustering(g, i):
    nbrs = g.neighbors(i)
    k = len(nbrs)
    if k < 2:
        return 0.0
    links = 0
    for x in range(k):
        for y in range(x + 1, k):
            a, b = nbrs[x], nbrs[y]
            if g.are_adjacent(a, b):        <--
                links += 1
    pairs = k * (k - 1) / 2                 <--
    return links / pairs
```

Nobody writes a breadth-first search, and nobody wraps igraph in a function either — the wrapper is two lines that are trivial once the API has been seen. Section 3 teaches `igraph.Graph(n=..., edges=...)` and `g.distances(source=s)` on the seven-person network, with cells to experiment in. Question 1(c) is then set as an empty cell: compute Ringville's three numbers with igraph and put them in three variables, marked against answers the notebook keeps hidden.

```
g = igraph.Graph(n=TOWN_N, edges=ring_edges(TOWN_N, TOWN_HALF))
counts = g.distances(source=0)[0]

answer_total   = sum(counts)
answer_average = sum(counts) / (TOWN_N - 1)
answer_worst   = max(counts)
```

The one thing that catches people is the `[0]`: `distances` accepts several sources, so it always returns a list of rows.

Run on the sheet's town they give **36 / 2.4 / 4** and **0.5** for every person, which are Question 1(c) and the darkened pairs in Part 3 again, out of code that never looked at the drawing.

Finished early — the number that lies. The notebook then measures $\sigma = (C/C_{\text{rand}})/(L/L_{\text{rand}})$ on a plain ring of *one thousand* people with **no shortcuts at all**, and gets $\sigma = 4.96$: a strong small world, by a test applied to a town that has none. Turn the n dial up and the verdict gets *stronger*, not weaker.

The four quantities, for a ring of n people with k friends each:

$$C = \tfrac{1}{2}, \qquad L \approx \tfrac{n}{8}, \qquad C_{\text{rand}} = \tfrac{k}{n}, \qquad L_{\text{rand}} \approx \tfrac{\ln n}{\ln k}$$

Where the two random-town numbers come from, since the notebook derives them and the question may come back. The random reference is the Erdős–Rényi graph $G(n, p)$: of the $\binom{n}{2}$ pairs who could be friends, each pair is made a friendship independently with the same probability p. Matching the average degree fixes p — person i has $n-1$ possible friends, each realised with probability p, so $k = p(n-1)$ and $p = k/(n-1)$, which is k/n for large n. For clustering: given that a and b are both friends of i, independence means the pair (a, b) is still a friendship with probability p, so $C_{\text{rand}} = p$. The expression contains no reference to i or to who i's friends are, which is what the independence assumption removed.

L_{rand} is the tree of 7(a). About k people are one handshake away and k^2 are two away, so $k^L \approx n$, and taking logs gives $L_{\text{rand}} \approx \ln n/\ln k$. Question 7(b) is exactly this derivation's false assumption: branches that double back onto people already in the tree make k^d an overestimate of the reach, so $\ln n/\ln k$ is an *under*estimate of the real distance. The shape is right even so, and the shape is the point — $\ln n$, not n. Eight billion people at $k = 150$ gives $\ln(8 \times 10^9)/\ln 150 \approx$ **4.5**, which is six degrees.

$$\frac{C}{C_{\text{rand}}} = \frac{n}{8}, \qquad \frac{L}{L_{\text{rand}}} = \frac{n}{8} \cdot \frac{\ln k}{\ln n}, \qquad \sigma = \frac{n/8}{(n/8)(\ln k/\ln n)} = \frac{\ln n}{\ln k}$$

The n cancels exactly. A ring is $n/8$ times more clustered than a random town *and* $n/8$ times further across, and σ divides one by the other, so it has measured nothing about this town at all. What survives the cancellation is the leftover $\ln n/\ln k$, which is there only because a random town's paths grow like $\ln n$ — so slowly that the ring's enormous path-length penalty is discounted by that factor, and the clustering side wins by it. Any network whose clustering does not fall with n scores $\sigma > 1$ automatically, shortcuts or no shortcuts. **σ was certifying that C is constant, not that shortcuts exist.**

The published repairs both add a *second* yardstick — a lattice, at the other end from the random graph — because "small world" was always a claim about the middle of a range, and σ only ever looked at one end:

$$\omega = L_{\text{rand}}/L - C/C_{\text{latt}} \qquad \text{(Telesford 2011; } -1 \text{ lattice, } 0 \text{ small world, } +1 \text{ random)}$$

At $n = 1000$, $k = 4$:

town	σ	ω	SWI
plain ring, $p = 0$	**4.96** passes	-0.96	0.00
rewired, $p = 0.05$	47.7	-0.41	**0.81** largest
rewired, $p = 0.2$	48.1	0.21	0.51
random, $p = 1$	0.47	0.93	0.00

The same network is a *strong small world* to σ and an *almost perfect lattice* to ω, and as n grows the two run in opposite directions. The price of the repair is that C_{latt} and L_{latt} have to come from somewhere: here the town *is* a lattice so they are free, and on real data you have to build a lattice with the same degrees yourself, which is not a small job. That is part of why σ is still the one people quote.

The whole notebook with every blank filled in is `lab-solutions.py`, built from `lab.py` by `tools/build_lab_notebooks.py m02-small-world`.